

Generation Interconnection System Impact Study Report

For Queue Project Q22-2026

Ansonia 138 kV

May 04, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2026. The Interconnection Customer proposes a 163.5 MW battery energy storage facility interconnecting at the Ansonia 138 kV substation in the New Jersey area. The request was received on March 07, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 668 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 21 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$18,383,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Crawfordsville Power Plant-Roxboro 138 kV: 147.5% loading (257 MVA) following the outage of the White Rock-Nike-Magnolia 1708 138 kV; DFAX = 8.1%.

For the loss of Locust Grove 345/138kV TX, the Getzville Substation-Fanwood 138 kV loads to 141.4% of its 224 MVA rating. Project DFAX: 14.9%.

Crawfordsville Power Plant-Pine Tree Acres 138 kV Line: 141.2% loading (686 MVA) following the outage of the Magnolia 1708-White Rock-Nike 138 kV; DFAX = 5.5%.

For the loss of Locust Grove 345/138kV Transformer, the Guion-Getzville 138 kV loads to 139.2% of its 217 MVA rating. Project DFAX: 14.9%.

For the loss of Locust Grove 345 kV/138 kV Autotransformer, the Guion-Jacaranda 138 kV loads to 138.2% of its 183 MVA rating. Project DFAX: 14.9%.

For the loss of White Rock-Nike-Magnolia 1708 138 kV, the Shredder-Roxboro 138 kV loads to 131.4% of its 324 MVA rating. Project DFAX: 8.1%.

Clearwater-Sweden Switches 138 kV: 129.2% loading (321 MVA) following the outage of the Ansonia 345 kV/138 kV TX; DFAX = 44.3%.

Ansonia-Fanwood 138 kV: 117.1% loading (302 MVA) following the outage of the Meadow Road-Hoist 138 kV line; DFAX = 21.3%.

Hoist-Meadow Rd 138 kV line: 114.3% loading (254 MVA) following the outage of the Ansona 345 kV/138 kV Transformer; DFAX = 30.7%.

For the loss of Ansonia 345 kV/138 kV Transformer Bank, the Hoist-Kilner 138 kV loads to 114.0% of its 253 MVA rating. Project DFAX: 30.7%.

For the loss of Wallenpaupack 500/345kV XFMR, the W Gallatin-Felida 345 kV loads to 110.9% of its 1,858 MVA rating. Project DFAX: 11.4%.

Station E Toluca-Locust Grove 138 kV line: 105.3% loading (358 MVA) following the outage of the Crawfordsville Power Plant-Pine Tree Acres 138 kV L; DFAX = 7.1%.

For the loss of Ansonia 345/138 kV XFMR, the New Dover-Meadow Rd 138 kV loads to 103.2% of its 243 MVA rating. Project DFAX: 30.7%.

Laurel Ave-Orange 345 kV: 102.7% loading (1,247 MVA) following the outage of the Orange 345/138 kV Transformer; DFAX = 11.2%.

The N Mullet River Substation-Murray Hill Solar 138 kV is overloaded to 102.2% of its 161 MVA emergency rating for the loss of the Ansonia 345-138 kV Transformer Bank. The project contributes 25.0% DFAX (40.9 MW) to this overload.

Locust Grove-Shredder 138 kV: 101.9% loading (486 MVA) following the outage of the Magnolia 1708-White Rock-Nike 138 kV; DFAX = 7.3%.

Talenenergy Martins Creek Llc Harwood-Felida 345 kV Line: 101.1% loading (1,745 MVA) following the outage of the Felida 500-345 kV Auto; DFAX = 9.3%.

New Dover Substation-San Gerardo 1342 138 kV Line: 101.0% loading (238 MVA) following the outage of the Ansonia 345/138 kV Autotransformer; DFAX = 30.7%.

The Burns-Coronado 345 kV is overloaded to 100.7% of its 1,655 MVA emergency rating for the loss of the Clear Springs 500/345 kV Auto. The project contributes 11.5% DFAX (18.8 MW) to this overload.

Hills Chapel-Wallenpaupack 345 kV L: 100.6% loading (1,817 MVA) following the outage of the West Gallatin-Walnut 345 kV; DFAX = 11.8%.

For the loss of El Paso-Ravenna 500 kV, the Lerdo-Tidd 500 kV loads to 100.4% of its 2,594 MVA rating. Project DFAX: 7.7%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Roxboro-Crawfordsville Power Plant 138 kV L	Rebuild line with higher-capacity conductor	\$1,937,000	13% (\$255,000)
Getzville Substation-Fanwood 138 kV	Rebuild line with higher-capacity conductor	\$2,667,000	30% (\$814,000)
Crawfordsville Power Plant-Pine Tree Acres 138 kV	Rebuild line with higher-capacity conductor	\$1,326,000	4% (\$49,000)
Guion-Getzville 138 kV	Rebuild line with higher-capacity conductor	\$1,952,000	30% (\$596,000)
Guion-Jacaranda 138 kV	Rebuild line with higher-capacity conductor	\$6,662,000	30% (\$2,034,000)
Roxboro-Shredder 138 kV	Rebuild line with higher-capacity conductor	\$6,133,000	13% (\$806,000)
Sweden Switches-Clearwater 138 kV line	Rebuild line with higher-capacity conductor	\$3,391,000	100% (\$3,391,000)
Ansonia-Fanwood 138 kV	Rebuild line with higher-capacity conductor	\$1,380,000	58% (\$803,000)
Hoist-Meadow Road 138 kV	Rebuild line with higher-capacity conductor	\$727,000	100% (\$727,000)
Kilner Substation-Hoist 138 kV	Rebuild line with higher-capacity conductor	\$665,000	100% (\$665,000)
W Gallatin-Felida 345 kV line	Rebuild line with higher-capacity conductor	\$35,076,000	2% (\$861,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sta E Toluca-Locust Grove 138 kV	Reconductor line	\$828,000	7% (\$56,000)
Meadow Road-New Dover 138 kV	Reconductor line	\$3,458,000	100% (\$3,458,000)
Laurel Ave-Orange 345 kV	Reconductor line	\$4,857,000	4% (\$214,000)
N Mullet River Substation-Murray Hill Solar 138 kV	Reconductor line	\$707,000	100% (\$707,000)
Locust Grove-Shredder 138 kV	Reconductor line	\$376,000	13% (\$50,000)
Talenenergy Martins Creek Llc Harwood-Felida 345 kV Line	Reconductor line	\$12,665,000	2% (\$261,000)
San Gerardo 1342-New Dover 138 kV	Reconductor line	\$1,540,000	100% (\$1,540,000)
Burns-Coronado 345 kV	Reconductor line	\$9,938,000	3% (\$267,000)
Hills Chapel-Wallenpaupack 345 kV L	Reconductor line	\$22,600,000	3% (\$609,000)
Lerdo-Tidd 500 kV	Reconductor line	\$15,754,000	1% (\$220,000)
	Total allocated to project		\$18,383,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.